

KIRAN TOMLINSON

🏠 Redmond, WA · ✉ kitomlinson@microsoft.com · 🌐 www.kirantomlinson.com · 📅 March 16, 2026

EMPLOYMENT

Senior Researcher <i>Microsoft Research, AI Interaction and Learning Group</i>	2024–
Visiting Instructor <i>Carleton College, Department of Computer Science</i>	2023

EDUCATION

PhD, Computer Science Cornell University <i>Advisor: Jon Kleinberg</i>	2019–2024
MS, Computer Science Cornell University	2019–2022
BA, Computer Science and Mathematics Carleton College <i>Summa Cum Laude, Distinction in Computer Science and in Mathematics</i>	2015–2019

INTERNSHIPS

Microsoft Research <i>Productivity and Intelligence Group, with Jennifer Neville</i>	Summer 2022
Microsoft <i>Office of Applied Research, with Longqi Yang and Mengting Wan</i>	Summer 2021
NASA Johnson Space Center <i>Spacecraft Software Engineering Branch, with Nathan Uitenbroek</i>	Summers 2016 & 2017

PUBLICATIONS

*Equal contribution, (α – β) alphabetical order

17. Exclusion Zones of Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
AAAI, 2026
16. Lost in Transmission: When and Why LLMs Fail to Reason Globally
Tobias Schnabel*, **Kiran Tomlinson***, Adith Swaminathan, and Jennifer Neville
NeurIPS (spotlight, 3%), 2025
15. When the Universe is Too Big: Bounding Consideration Probabilities for Plackett–Luce Rankings
Ben Aoki-Sherwood, Catherine Bregou, David Liben-Nowell, **Kiran Tomlinson**, and Thomas Zeng (α – β)
AISTATS, 2025
14. Replicating Electoral Success
Kiran Tomlinson, Tanvi Namjoshi, Johan Ugander, and Jon Kleinberg
AAAI, 2025
13. Computational Perspectives on Individual and Collective Decision-Making
Kiran Tomlinson
PhD Dissertation, 2024
12. The Moderating Effect of Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
AAAI, 2024
11. Graph-Based Methods for Discrete Choice
Kiran Tomlinson and Austin R. Benson
Network Science 12(1), 2023
10. Workplace Recommendation with Temporal Network Objectives
Kiran Tomlinson, Jennifer Neville, Longqi Yang, Mengting Wan, and Cao Lu
KDD, 2023

9. Targeted Training for Multi-organization Recommendation
Kiran Tomlinson, Mengting Wan, Cao Lu, Brent Hecht, Jaime Teevan, and Longqi Yang
ACM Transactions on Recommender Systems 1(3), 2023
8. Ballot Length in Instant Runoff Voting
Kiran Tomlinson, Johan Ugander, and Jon Kleinberg
 AAAI, 2023
7. Choice Set Confounding in Discrete Choice
Kiran Tomlinson, Johan Ugander, and Austin R. Benson
 KDD, 2021
6. Learning Interpretable Feature Context Effects in Discrete Choice
Kiran Tomlinson and Austin R. Benson
 KDD, 2021
5. Choice Set Optimization Under Discrete Choice Models of Group Decisions
Kiran Tomlinson and Austin R. Benson
 ICML, 2020
4. Summarizing Diverging String Sequences, with Applications to Chain-Letter Petitions
 Patty Commins, David Liben-Nowell, Tina Liu, and **Kiran Tomlinson** (α - β)
 31st Annual Symposium on Combinatorial Pattern Matching (CPM), 2020
3. Distance Measures for Tumor Evolutionary Trees
 Zach DiNardo*, **Kiran Tomlinson***, Anna Ritz, and Layla Oesper
Bioinformatics 38(7), 2020
2. Parameter, Noise, and Tree Topology Effects in Tumor Phylogeny Inference
Kiran Tomlinson and Layla Oesper
BMC Medical Genomics 12(10), 2019
1. Examining Tumor Phylogeny Inference in Noisy Sequencing Data
Kiran Tomlinson and Layla Oesper
 IEEE Int. Conf. on Bioinformatics and Biomedicine (BIBM), 2018

PREPRINTS

2. Reasoning about Reasoning: BAPO Bounds on Chain-of-Thought Token Complexity in LLMs
Kiran Tomlinson, Tobias Schnabel, Adith Swaminathan, and Jennifer Neville
 arXiv, 2026
1. Working with AI: Measuring the Applicability of Generative AI to Occupations
Kiran Tomlinson, Sonia Jaffe, Will Wang, Scott Counts, and Siddharth Suri
 arXiv, 2025

PATENTS

2. Intelligent system and method of optimizing cross-team information flow
 Longqi Yang, Mengting Wan, Cao Lu, Jennifer Neville, and **Kiran Tomlinson**
 US20240354703A1 (pending), filed Apr 20, 2023
1. Targeted training of inductive multi-organization recommendation models for enterprise applications
Kiran Tomlinson, Longqi Yang, Mengting Wan, Cao Lu, Brent Hecht, and Jaime Teevan
 US11983649B2, May 14, 2024 (filed Oct 26, 2021)

INVITED TALKS

- “Exclusion Zones of Instant Runoff Voting.” COMSOC Video Seminar, Feb 2026.
- “Working with AI: Measuring the Applicability of Generative AI to Occupations.” Louisiana State Board of Elementary and Secondary Education, Sep 2025.
- “Algorithmic Perspectives on Instant Runoff Voting.” Cornell CS Theory Seminar, Aug 2023.

“Choices in Networks.” NetSci–SINM, Jul 2023.

“A Computational Look at Ranked-Choice Voting.” Carleton College CS Tea, May 2023.

“Choice Models for Networks and Networks for Choice Models.” Network Dynamics and Choice Theory Workshop @ UVM, May 2022.

“Feature Context Effects and Choice Set Confounding in Discrete Choice.” WPI, CS 525, Apr 2021.

“Feature Context Effects and Choice Set Confounding in Discrete Choice.” Cornell/MIT MURI, Mar 2021.

“Learning Interpretable Feature Context Effects in Discrete Choice.” Stanford Ugander Lab, Dec 2020.

WORKSHOP TALKS

“Working with AI: Measuring the Applicability of Generative AI to Occupations.” NABE Tech Economics Conference (Plenary), Nov 2025.

“Working with AI: Measuring the Applicability of Generative AI to Occupations.” Wharton Business & Generative AI Conference, Sep 2025.

“Exclusion Zones in Instant Runoff Elections.” AMS Special Session on Mathematics of Decisions, Elections, and Games @ JMM, Jan 2025.

“Workplace Recommendation with Temporal Network Objectives.” NetSci, Jul 2023.

“Learning Context Effects in Triadic Closure.” NetSci–SINM, Sep 2020.

“Distance Measures for Tumor Evolutionary Trees.” RECOMB–CCB (with Zach DiNardo), May 2019.

AWARDS

The Web Conference Outstanding Reviewer	2024
Phi Beta Kappa	2019
Sigma Xi	2019
CRA Outstanding Undergraduate Researcher Honorable Mention	2018
Goldwater Scholarship Honorable Mention	2018
Patricia V. Damon Scholarship	2018
NSF Student Travel Award (to BIBM)	2018
Carleton College Dean’s List	2016, 2017, 2018
National Merit Scholarship	2015–2018

TEACHING

Carleton College (Instructor)

CS 201: Data Structures	Winter 2023 (×2), Spring 2023
CS 202: Mathematics of Computer Science	Spring 2023

Cornell University (Teaching Assistant)

CS 2110: Object-Oriented Programming and Data Structures	Fall 2019
CS 2800: Discrete Structures	Spring 2020

Carleton College (Prefect)

CS 201: Data Structures	Winter 2018, Spring 2019
CS 202: Mathematics of Computer Science	Fall 2017, Spring 2018
CS 254: Computability and Complexity	Fall 2018, Winter 2019

SERVICE

Workshop Program Chair

What Can(t) Transformers Do? @ NeurIPS ’25

The Workshop on Human and Machine Decisions @ NeurIPS ’21

Program Committee

AAAI ’23, ’24, ’25, ’26

AISTATS ’25, ’26

ICLR '26
ICML '25, '26
ICWSM '25
NeurIPS '24, '25
WebConf '24, '25, '26

Journal Reviewer

Science Advances
Biometrika
Data Mining and Knowledge Discovery
Annals of Mathematics and Artificial Intelligence
Information Processing and Management

PhD Admissions Committee

2021, 2022

Cornell University Computer Science Department

Graduate and Professional Student Assembly Representative (elected)

2019–2020

Cornell Computer Science Graduate Organization

CERTIFICATIONS

Private Pilot (ASEL), FAA	2026
Amateur Radio Operator (Technician), FCC	2017
Open Water Diver, PADI	2015